**PELAYO PACHE RODRÍGUEZ**

**INDUSTRIAL ELECTRONICS & AUTOMATION ENGINEERING STUDENT**

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## Education

**UNIVERSIDAD DE OVIEDO (EPI Gijón)** **Gijón, Spain**

*B.Sc. in Industrial Electronics and Automation Engineering Sept. 2022 – Present (Expected Jun. 2027)*

- **Academic Record:** Cumulative GPA: **7.49 / 10.0** (Equivalent to UK 2:1 Honours / ∼3.2 US GPA).
- **Bachelor's Thesis (TFG):** Currently developing TFG in direct collaboration with **Universidad de León** in **Aeronautical Engineering**.
- **Relevant Coursework: Digital Electronics**, **Embedded Systems**, **Microelectronics**, **Power Electronics**, **Control Engineering**, Analog Circuit Design, Electronic Instrumentation, Industrial Automation.

**LIVERPOOL JOHN MOORES UNIVERSITY (LJMU)** **Liverpool, United Kingdom**

*Academic Exchange Year – Faculty of Engineering and Technology* *Sept. 2025 – Jun. 2026*

- **Academic Classification:** Awarded **Distinction / First Class Honours** (Equivalent GPA in the Spanish academic system: **9.15 / 10.0**).
- Full academic year taught entirely in English, focusing on advanced embedded automation, sensor integration, applied control theory, and industrial systems within an international UK setting.

**SANTO DOMINGO DE GUZMÁN COLLEGE (Dominicos)** **Oviedo, Spain**

*High School Diploma (Scientific-Technological Track)* *Sept. 2020 – May 2022*

## Engineering Experience & Technical Projects

**RESEARCH COLLABORATION GRANT** **Oviedo / Gijón, España**

*Selected for Top Research Collaboration Project* *Sept. 2026 – Present*

- **Innovation Selection:** Development of laboratory test equipment for satellite constellation release testing or solar panel deployment in simulated microgravity.
- **Project Defense:** Delivery of a comprehensive project report for formal defense, endorsed by the faculty advisor for submission to the Ministry of Education.

**ART – ASTURIAS RURAL TECH (UNIOVI EMPRENDE)** **Oviedo / Gijón, España**

*Co-Fundador y Desarrollo Tecnológico (Top 12 Proyectos de Innovación)* *Sept. 2026 – Presente*

- **Top Innovation Finalist Selection:** Selected as **Finalists** among the **Top 12 Innovation Projects** in the official entrepreneurship program *Uniovi Emprende*, organized in strategic partnership with **Banco Santander** and leading regional business entities; qualified for and defended the initiative before an executive evaluation tribunal on **November 5, 2026**.
- **Direct Booking Platform & Digital Architecture:** Co-designed and deployed the web platform architecture and proprietary direct booking engine (ART Portal) engineered to eliminate abusive 15%–25% commissions from multinational OTAs (Booking, Airbnb), enabling rural accommodations and commerce to retain 100% of their net margin.
- **Field Methodology («Tú a Tú») & Commercial Pilot Validation:** Spearheaded on-the-ground technical consulting and local SEO diagnostics in rural municipalities; validated the model through an operational pilot with a traditional enterprise in Pola de Laviana, achieving a **4× increase in direct customer contacts and calls** via Google Maps optimization.
- **Strategic Financing & Market Modeling:** Structured a 4-tier scalable monetization model ad-

dressing a **EUR 22M TAM** (2,500+ rural businesses in Asturias), lowering client adoption barriers to **EUR 0 upfront cost** by channeling official *Kit Digital* subsidies (EUR 3,000–EUR 12,000 grants) to foster long-term rural business sustainability.

**UNIOVI ETECH-RACING (FORMULA STUDENT)** **Gijón, Spain**
*Low Voltage Hardware & Electronics Engineer* *Sept. 2024 – May-2026*

- **PCB Design:** Designed, routed, and validated multi-layer printed circuit boards (PCBs) using **Altium Designer** (Advanced Level) for the vehicle's Low Voltage (LV) electrical architecture, ensuring high signal integrity and strict compliance with Formula Student safety regulations.
- **Industrial Communications:** Implemented and debugged **CAN bus communication** protocols (500 kbps) for real-time data exchange between the Vehicle Control Unit (ECU), telemetry unit, and Battery Management System (BMS).
- **Wiring  Bench Testing:** Engineered custom modular wiring harnesses with motorsport connectors and EMI shielding. Bench testing and **tin soldering** (THT  SMD).

**UNIOVI INNOVATION SKILLS** **Gijón, Spain**
*Technical Lead & Hardware Prototyping (Winner – Most Innovative Project)Academic Year 2023 – 2024*

- Spearheaded technical design of an automated industrial monitoring solution developed in partnership with municipal innovation hub *Gijón Impulsa* and elevator manufacturer *Tresa Ascensores*.
- Engineered sensor telemetry interface and predictive logic; presented functional prototype to an executive jury, winning **1st Prize for Most Innovative Project**.

## Additional Work Experience & Leadership

**PRIVATE TUTOR IN MATHEMATICS & PHYSICS** **Oviedo, Spain**
*Private Tutor* *2022 – Present*

- Delivered personalized instruction in calculus, linear algebra, and classical mechanics to over 10 secondary and pre-university students, strengthening technical communication skills.

**EDP (ELECTRICIDADE DE PORTUGAL) – FIDMA TRADE FAIR** **Gijón, Spain**
*FIDMA EDP Assistant* *August 2023*

- Customer support, brand representation, and public engagement on renewable energy and transition projects at EDP's fair pavilion.

**ALVAPAN** **Oviedo, Spain**
*Delivery Assistant (Summer Contract)* *Summer 2022*

- Managed delivery logistics, route organization, and customer service under tight schedules in a demanding environment.

## Technical & Social Skills

- **Systems Engineering & MBSE (HIGHLIGHTED): SysML v2** – Model-Based Systems Engineering (MBSE), with practical application both in specialized **Dassault CATIA / MagicDraw** software and in agnostic model architectures **without CATIA** (continuously advancing learning).
- **Hardware & EDA Design: Altium Designer** (Advanced), KiCAD, Multilayer PCB Design, Schematic Capture, Circuit Prototyping, **Tin Soldering** (THT  SMD), Bench Equipment (Oscilloscopes, Logic Analyzers, Regulated Power Supplies).
- **Embedded Systems & Logic:** Microcontrollers (**STM32**, **ARM Cortex**, **Microchip PIC** with **MPLAB**), **VHDL (Xilinx Vivado)**, **CAN Bus**, SPI, I2C, UART, Digital Logic, Siemens PLC (**TIA Portal**).
- **Simulation & Modeling: MATLAB & Simulink**, **PLECS**, **PSIM**, AutoCAD.
- **Programming Languages: C/C++**, **Python**, **Java**, **R**.

- **Social Management Skills:** Leadership, Teamwork in demanding technical environments, Problem-solving and management skills, Technical issue analysis and resolution, Adaptability and quick learning.
- **Languages:**

  **Spanish:** Native / Bilingual.

  **English: C1 Full Professional Fluency** (Oxford ELLT C1; Cambridge B2; Cambridge IELTS C1 ).

  **French:** Basic / Very limited proficiency.

## Certifications & Training

- **Cambridge IELTS (C1 Proficient):** University of Cambridge / British Council.
- **Oxford ELLT (C1 Proficient):** University of Oxford / OI Digital Institute.
- **British Airways – Engineering Job Simulation:** Aircraft mechanics, systems safety, and fleet maintenance analysis (Forage).
- **Santander Open Academy:** Artificial Intelligence & Large Language Models with Gemini.
- **Cambridge First Certificate in English (FCE - B2):** University of Cambridge.
- **Driving License:** Class B.